

Machine Learning Primer - Document 2

Section 1: Supervised vs Unsupervised Learning

Machine learning is categorized into three primary paradigms:

1. Supervised Learning: Algorithms are trained on labeled data where inputs are paired with target outputs.

Common supervised tasks include classification and linear regression.

2. Unsupervised Learning: Algorithms discover hidden patterns, groupings, or clusters in unlabeled data.

Prominent unsupervised techniques include k-means clustering and principal component analysis (PCA).

3. Reinforcement Learning: Agents learn through trial-and-error by interacting with an environment to maximize reward.

Section 2: Gradient Descent and Optimization

Optimization is the engine of modern neural network training. Gradient descent is an iterative first-order optimization algorithm used to find a local minimum of a differentiable objective function. The algorithm calculates the gradient of the loss function with respect to model weights and updates the weights in the opposite direction of the gradient scaled by a hyperparameter called the learning rate.

Variants such as Stochastic Gradient Descent (SGD) and Adam are widely used in practice.